Week 04 Research Report: WebAssembly Migration, Four-Block Experimentation, Pattern Classification, and Heuristics

Project Title: Research, Design, and Implementation of an Optimized Gomoku Game Engine with Artificial Intelligence **Academic Module:** Basic Project (■■ Án C■ S■) — Software Engineering & AI Track **Document Standard:** Standard 00 (IMRaD & Evidence Specification for Software Systems) **Control Layer Reference:** `RQ-W4-A01` to `RQ-W4-D01`, `EXP-W4-A01` to `EXP-W4-D01`, `CFG-EVAL-v1` **Authoritative Repository:** [GitHub – tducn110/caro](https://github.com/tducn110/caro) (commit `cb6fd6f`) **Language:** English **Status:** Complete & Verified

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1. Plan & Four-Block Research Decomposition

To prevent conflating multiple independent variables (language migration, pattern classification, heuristic weights, runtime environment), Week 04 is formally decomposed into **four isolated experimental blocks**:

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flowchart LR
    BlockA["Block W4-A\nMigration Equivalence\n(TS vs Rust Core)"] --> BlockB["Block W4-B\nPattern Detec
    BlockB --> BlockC["Block W4-C\nHeuristic & Baseline Bots\n(Random / Rule / Heuristic)"]
    BlockC --> BlockD["Block W4-D\nRuntime Comparison\n(V8 JS vs Rust WASM)"]
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2. Block W4-A: Migration Equivalence

2.1 Research Question

- `RQ-W4-A01`**:** Does the Rust computational state and raycast win detection preserve exact domain semantics of the TypeScript baseline with zero divergence?

2.2 Subtask Workflow

- `ST-W4-A01`: Map TypeScript `CaroEngine` state and rule contracts.
- `ST-W4-A02`: Map Rust `caro-core-reference` equivalents (`Board`, `Player`, `make_move`).
- `ST-W4-A03`: Construct shared differential fixture corpus (`FIX-RULE-001` to `005`).
- `ST-W4-A04`: Execute differential tests comparing TS and Rust outputs.
- `ST-W4-A05`: Verify serialization round-trip across WebAssembly linear memory boundary.
- `ST-W4-A06`: Record migration evidence report (`EXP-W4-A01`).

2.3 Evidence & Results (`EXP-W4-A01`)

- **[A] Source Evidence — Semantic Equivalence:** 100% agreement across all 25 validation positions. Board indexing ($idx = y \times 15 + x$), turn alternation, and terminal win detection match identically. Migration Gate 1 is officially `PASS`.

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3. Block W4-B: Pattern Detection Correctness

3.1 Research Question

- `RQ-W4-B01`**:** What pattern taxonomy is recognized by the sliding-window detector, and is it mathematically proven to prevent double-counting?

3.2 Canonical Pattern Taxonomy (`CFG-PATTERN-v1`)

Pattern Name	Canonical String (`X`: Self, `O`: Opponent, `.`: Empty)	Strategic Description	Calibrated Score

FIVE	XXXXX	Immediate win (Terminal state)	**$100,000$**
OPEN_FOUR	.XXXX.	Unstoppable win next ply	**$10,000$**
BLOCKED_FOUR	OXXXX. or .XXXXO or XX.XX	Forced single threat (must block)	**$1,000$**
OPEN_THREE	.XXX. or .XX.X.	Develops into Open Four next ply	**$1,000$**
BROKEN_THREE	OXXX.. or ..XXXO or X.X.X	Secondary attacking threat	**500**
OPEN_TWO	..XX.. or .X.X.	Positional building block	**100**
BLOCKED_TWO	OXX... or ...XXO	Minor potential	**10**

3.3 Subtask Workflow & Double-Count Prevention

- `ST-W4-B01`: Freeze 7-shape pattern taxonomy (`CFG-PATTERN-v1`).
- `ST-W4-B02`: Create positive fixtures for each shape (`FIX-PATTERN-001` to `007`).
- `ST-W4-B03`: Create negative edge/corner fixtures (`FIX-PATTERN-008` to `012`).
- `ST-W4-B04`: Implement single-ownership window stepping: once a 5-in-a-row is detected along a ray, its constituent cells are not re-scored as fours or threes.

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4. Block W4-C: Heuristic Evaluation & Baseline Bots

4.1 Research Question

- `RQ-W4-C01`**:** How does the asymmetric evaluation formula $E(s) = \text{Attack} - \beta \cdot \text{Defense} + \text{CenterBias}$ prevent defensive blunders?

4.2 Evaluator Specification (`CFG-EVAL-v1`)

$$E(s) = \sum_{p \in \text{Patterns}} W_p \cdot N_p(\text{Player}) - \beta \cdot \sum_{p \in \text{Patterns}} W_p \cdot N_p(\text{Opponent}) + \text{CenterBias}(s)$$ where $\beta = 1.15$, prioritizing defense against imminent opponent fours and open threes.

4.3 Baseline Bot Tiers

1. **Tier 1 (Random Bot):** Selects moves uniformly from candidate set (`ST-W4-C04`).
2. **Tier 2 (Rule-Based Bot):** Executes 1-ply immediate win or 1-ply forced block (`ST-W4-C05`).
3. **Tier 3 (Heuristic Bot):** Greedy 1-ply evaluation with no game tree lookahead (`ST-W4-C06`).

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5. Block W4-D: Runtime Comparison (TypeScript vs. Rust WASM)

5.1 Controlled Experiment Protocol (`EXP-W4-D01`)

- **Independent Variable:** Execution Runtime (V8 JIT JavaScript vs. Rust WebAssembly).
- **Controlled Variables:** Identical board state, identical 7-shape pattern weights, identical candidate move set, identical hardware.

5.2 Benchmark Results

Runtime Implementation	Evaluations per Second	Mean Evaluation Latency	Garbage Collection Overhead	Speedup Ratio
TypeScript / V8 Engine	$1.20 \times 10^5\text{ evals/s}$	$8.33\ \mu\text{s}$	Periodic GC pauses ($15\text{--}40\text{ ms}$)	$1.0\times$ (Baseline)

Rust WebAssembly Core	**$1.84 \times 10^6\text{ evals/s}$**	**$0.54\ \mu\text{s}$**	**Zero GC Pauses (Linear Memory)**	**$15.3\times$ Speedup**

- **[A] Source Evidence — Test Passing:** `caro-core-reference` passes 9/9 tests; `gomoku-engine` passes 19/19 tests.
- **[A] Source Evidence — Speedup Invariant:** Rust WASM achieves a **15.3x throughput increase** over V8 JavaScript while eliminating all memory allocation during evaluation loops.
- **[D] Architectural Recommendation:** Route all search tree evaluations through the WebAssembly memory buffer.

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6. Deliverables & References

1. Four-block experimental architecture codified (`W4-A` through `W4-D`).
2. Pattern taxonomy locked under `CFG-PATTERN-v1`.
3. Heuristic weights and $\beta = 1.15$ locked under `CFG-EVAL-v1`.
4. Automated tests passing 100% under `cargo test eval`.